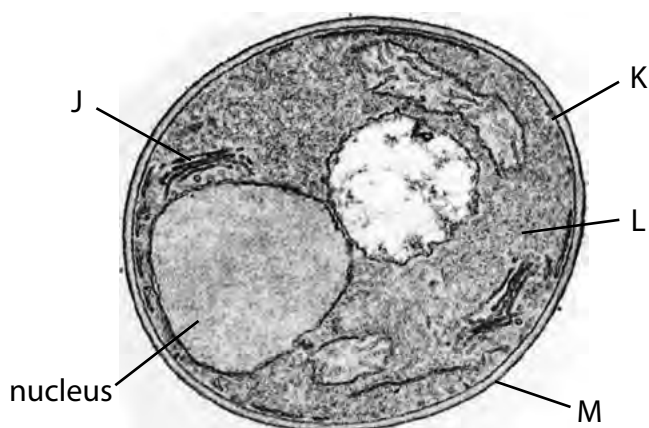
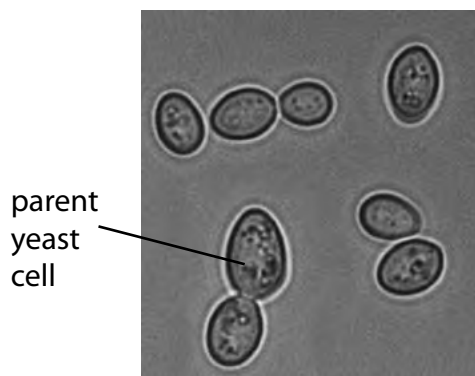


- 3 The photographs show yeast cells, seen using a light microscope and an electron microscope.



Yeast cells seen using a light microscope

Yeast cell seen using an electron microscope

Used under CC License from: https://commons.wikimedia.org/wiki/File:Zygosaccharomyces_bailii_cells.jpg

- (a) Which structure identifies yeast as a eukaryotic organism?

(1)

- ☐ A J
- ☐ B K
- ☐ C L
- ☐ D M

- (b) Explain why structure J can be seen using the electron microscope but not the light microscope.

(2)

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- (c) Explain why the nuclear envelope cannot be seen as two membranes using this electron microscope.

(2)

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- (d) Yeast cells reproduce asexually by a process called budding.

The parent yeast cell produces a bud.

- (i) Explain the importance of mitosis in budding.

(3)

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- (ii) Once the bud is large enough, it separates from the parent yeast cell.

The rate at which budding happens depends on the availability of oxygen and nutrients.

Suggest why the availability of oxygen and nutrients determines the rate of budding.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 3 = 11 marks)